

Name:

Student ID:

CONVEXITY PROBLEM SHEET

1

(a) Starting from the definition of a convex function

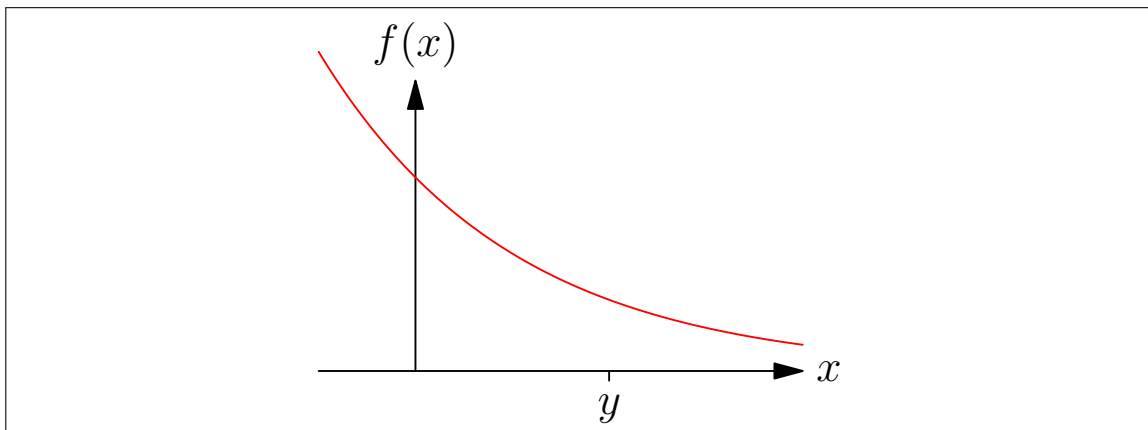
$$f(ax + (1 - a)y) \leq af(x) + (1 - a)f(y) \quad (1)$$

Let $a = \epsilon/(x - y)$ and rearrange the inequality to give

$$(x - y) \left(\frac{f(y + \epsilon) - f(y)}{\epsilon} \right)$$

on the left-hand side. Taking the limit $\epsilon \rightarrow 0$ show that the function $f(x)$ lies above the tangent line $t(x) = f(y) + (x - y)f'(y)$ going through the point y .
[4 marks]

(b) Sketch the tangent line, $t(x)$, at the point y in the graph shown below. [1 mark]



1

(c) Starting from the inequality for a convex function

$$f(x) \geq f(y) + (x - y)f'(y) \quad (2)$$

consider the case $y = x + \epsilon$, then by Taylor expanding $f(x + \epsilon)$ and $f'(x + \epsilon)$ around x and keeping all terms up to order ϵ^2 show that for a convex function $f''(x) \geq 0$. [4 marks]

4

(d) Prove that $-\log(x)$ is convex-up for $x > 0$.

[1 mark]

1

End of question 1

(a) $\frac{4}{4}$ (b) $\frac{1}{1}$ (c) $\frac{4}{4}$ (d) $\frac{1}{1}$ Total $\frac{10}{10}$

2

- (a) If $\|x\|$ is a proper norm use the triangular inequality ($\|x + y\| \leq \|x\| + \|y\|$), linearity of a norm ($\|ax\| = a\|x\|$) and the definition of convexity, to show that the norm is convex. [5 marks]

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5

- (b) Consider a classification problem where $\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})$ is the probability that a learning machine with parameters $\boldsymbol{\theta}$ predicts that input $\mathbf{x}$ belongs to class $c \in \mathcal{C}$. Assume the training is stochastic so the probability of obtaining parameters $\boldsymbol{\theta}$ is $\rho(\boldsymbol{\theta})$. Let $\hat{m}_c(\mathbf{x}) = \mathbb{E}_{\boldsymbol{\theta}}[\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})]$ be the output of the mean machine for class c . Assuming that for a data point $(\mathbf{x}, y)$, where y is a class label, we use a cross entropy loss

$$L(\mathbf{x}, y, \boldsymbol{\theta}) = - \sum_{c \in \mathcal{C}} \mathbb{I}[y = c] \log(\hat{f}_c(\mathbf{x}|\boldsymbol{\theta})),$$

show that the expected loss over inputs and parameters can be written as the expected loss of the mean machine plus a second loss. Use Jensen's inequality ($\mathbb{E}[\log(X)] \leq \log(\mathbb{E}[X])$) to show the second term is positive. [5 marks]

5

End of question 2

(a) $\frac{\quad}{5}$ (b) $\frac{\quad}{5}$ Total $\frac{\quad}{10}$